

# Structural Recursive Self-Improvement

DBM-SI as a Rich Evaluator and Improvement Infrastructure - SRSI v1.0.0

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## Overview

Structural Recursive Self-Improvement (SRSI) presents a research framework for treating recursive self-improvement not merely as recursive candidate generation under increasing compute, but as a governed structural process built on Rich Evaluators, Counter-Evidence, structural comparison, localization, verification, cumulative improvement memory, and selective promotion. It introduces the Evaluator Bottleneck hypothesis, develops Localized and Per-Node RSI as alternatives to whole-system self-replacement, and maps DBM-SI structures into one concrete evaluator and improvement infrastructure.

***"Compute determines how hard RSI can search. Evaluators determine what RSI learns to become."***

## Core Structural Loop

|          |          |                  |          |        |        |      |
|----------|----------|------------------|----------|--------|--------|------|
| Generate | Evaluate | Counter-Evaluate | Localize | Verify | Govern | Fold |
|----------|----------|------------------|----------|--------|--------|------|

## Five Pillars

**Rich Evaluation** - determine what actually improved and under which context. **Counter-Evidence** - actively search for why an apparent improvement may fail. **Localization** - improve structures without blindly replacing the whole system. **Structural Memory** - preserve validated and rejected improvement experience. **Improvement Governance** - separate validation, promotion, deployment, and action authority.

## Key Contributions of v1.0.0

**Evaluator Bottleneck:** as candidate generation becomes abundant, trustworthy machine-operable judgment may become a critical constraint. **Anti-Goodhart RSI:** Two-Way CCC and Counter-Evidence Search provide mechanisms for structural A/B comparison and falsification. **Localized RSI:** recursive improvement can operate through Per-Node Intelligence, branching, Leftover states, rollback, and selective promotion. **AI-SI-RSI Gold Rush:** Rich Evaluators, Structural Search, Evaluator Packs, Improvement Memory, and Improvement Runtimes are proposed as a possible next engineering frontier.

## Release Contents

Six core research papers: SRSI-001 through SRSI-006; five canonical figures; README; START-HERE; CONTENTS; FIGURE-INDEX; GLOSSARY; FUTURE-DIRECTIONS; CHANGELOG; citation metadata; and GitHub Release Notes.

## Scope

SRSI v1.0.0 is a **research framework, structural proposal, engineering hypothesis, and research direction**. It does not claim a completed autonomous RSI system, a production self-improvement runtime, a solved Goodhart problem, or a guaranteed intelligence explosion. The next phase emphasizes Minimum Engineering Tests, beginning with AI Coding SRSI, Rich Evaluator benchmarking, Counter-Evidence runtimes, Localized RSI, and Improvement Memory.

## Links

**GitHub Repository:** <https://github.com/sizhet/Structural-Recursive-Self-Improvement>

**DOI:** To be added after the SRSI v1.0.0 Zenodo DOI is issued.

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**Canonical thesis:** Recursive self-improvement is not merely recursive generation under increasing compute. It is recursive generation under rich evaluation, counter-evidence, structural localization, cumulative memory, verification, and governed promotion.